



**Politecnico
di Torino**

ICT ENGINEERING FOR SMART SOCIETIES

COURSE: ADVANCED MACHINE LEARNING FOR IMAGE PROCESSING

Laboratory 4 Report

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1 Diffusion Models

Abstract

This report summarizes the experiments conducted in the Diffusion Models Lab, focusing on the use of Stable Diffusion for text-to-image generation, understanding its components, visualizing the diffusion process, and exploring advanced features like Classifier-Free Guidance and Image-to-Image translation.

2 Stable Diffusion Text-to-Image Generation

The first experiment uses a text prompt to generate an image. The main parameter of interest is `num_inference_steps`, which controls how many denoising iterations the model performs. A higher number of inference steps generally improves image fidelity because the diffusion process has more opportunities to refine the latent representation. However, more steps also increase computation time.

The text prompt used is "a horse that is running in the far west." We generate three images, each with a different `num_inference_steps` to underline the differences.



Figure 1: Images generated with `num_inference_steps` of 10, 25, 50 and 75

3 Model Components: UNet and Text Encoder

We inspect the two main pipeline components: `pipe.unet` and `pipe.text_encoder`.

- **UNet:** This is the denoising network. It receives the current noisy latent representation and conditioning information, then predicts the noise to remove at each step. In Stable Diffusion, the UNet is responsible for the majority of the image generation work.
- **Text encoder:** This module converts the text prompt into embeddings. These embeddings guide the UNet so that the generated image matches the prompt semantics. In this notebook, a CLIP-based text encoder is used.

4 Visualizing the Diffusion Process

A callback function captures intermediate images during generation by decoding latents every five diffusion steps. This reveals how the model transitions from noise to a final image. The diffusion process is iterative. By inspecting images at intervals, we can see how structure gradually appears and how the model refines details. This makes the denoising process tangible and helps explain why more steps often improve quality.

Key implementation choices:

- `callback_on_step_end`: used to execute code after each denoising step.
- `step % 5 == 0`: saves only every fifth step to reduce the number of snapshots while still showing a meaningful progression.
- Decoding latents to RGB images requires careful scaling and device handling, since the latents are stored in a special numeric format.

The saved images should show a transition from random noise to a more coherent horse image. Early frames are abstract; later frames increasingly reflect the final scene.

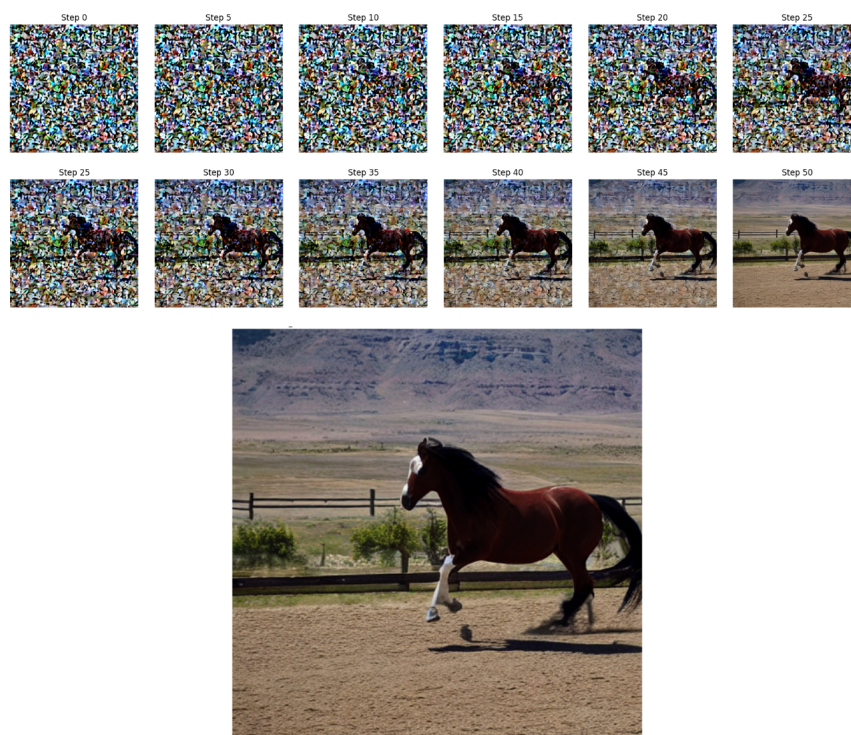


Figure 2: Snapshot sequence from the diffusion process.

5 Classifier-Free Guidance

Classifier-Free Guidance (CFG) adjusts how strongly the model follows the prompt. It balances between faithful prompt adherence and creative diversity.

The notebook evaluates three values:

- `guidance_scale = 1.0`: weak guidance, more freedom, less precise adherence to the prompt.
- `guidance_scale = 7.5`: typical default value, a reasonable tradeoff between prompt fidelity and image quality.
- `guidance_scale = 20.0`: strong guidance, produces a very prompt-specific image but can create unnatural artifacts or over-saturated details.

At low values, the model leans toward unconditional generation, so the output may be more ambiguous but often smoother. At high values, it is forced to honor the prompt very closely, which may emphasize textual details at the cost of naturalness.



Figure 3: Images generated with guidance scales of 1.0, 7.5, and 20.0.

6 Image-to-Image Translation

The final experiment uses the image-to-image pipeline to transform a horse painting into a zebra. The main parameter is `strength`, which controls how much of the original image is preserved.

Unfortunately, it was not possible to produce the transformed image due to memory issues. We have tried to use the suggested commands to unlock more GPU RAM, but the progress bar was constantly stuck: it did not work.

7 Conclusion

The lab demonstrates both basic and advanced aspects of Stable Diffusion. The key takeaway is that parameter choices directly affect output quality, prompt fidelity, and the amount of transformation.

- `num_inference_steps`: more steps generally improve details but cost more time.
- `guidance_scale`: controls strictness of prompt adherence versus creative freedom.
- `strength`: controls how much of the original image is preserved during image-to-image translation.